

Answer all questions.

1. (a) The table below shows the relationship between length, surface area and volume for three different organisms.

Organism	Length	Surface area (m ²)	Volume (m ³)	Surface area to volume ratio
amoeba	1µm	6 × 10 ⁻¹²	1 × 10 ⁻¹⁸	6000000 : 1
housefly	10mm	6 × 10 ⁻⁸	1 × 10 ⁻¹²	
dog	1m	6 × 10 ⁰	1 × 10 ⁰	6 : 1

The housefly is an organism adapted to a terrestrial mode of life, possessing an internal tracheal system for gas exchange.

- (i) Calculate the surface area to volume ratio of the housefly. [1]

Surface area to volume ratio =

- (ii) Give **one** advantage and **one** disadvantage of the tracheal system. [2]

Advantage:

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Disadvantage:

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- (b) Use the table to explain why an amoeba does not need a specialised gas exchange surface or circulatory system. [3]

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(c) State how mammals maintain the concentration gradient at their gas exchange surface. [2]

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(d) Suggest why cellular demand for oxygen is lower in a fish compared to a dog of the same size. [2]

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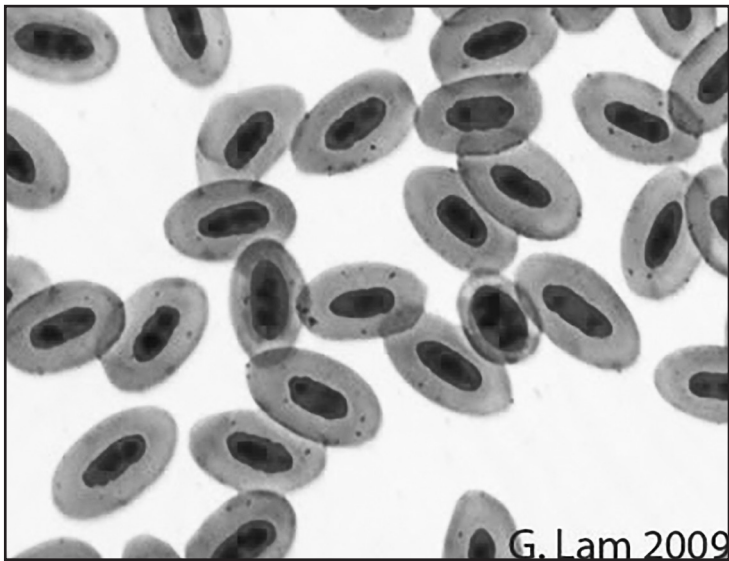
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- (e) Mammals evolved in the Triassic period when levels of oxygen in the atmosphere were 50% lower than now. Birds evolved much later, in the Jurassic period, when oxygen levels in the atmosphere approached present day levels.

The red blood cells of birds are elliptical in shape and have a nucleus.

Mammalian red blood cells are biconcave in shape, do not have a nucleus and are much smaller than the red blood cells of birds.



Red blood cells from birds

How do you account for the difference in size and structure of bird and mammalian red blood cells? [4]

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